Laws of exponents

1

a a^{s+t} **b** a^{s-t} **c** a^{2s} **d** $2a^{s}$

e $a^{s}+a^{t}$ **f** 2^{4} **g** $24y^{4}$ **h** $24x^{7}$

2

a $\dfrac{x^{2}}{4}$ **b** p^{5} **c** n^{7} **d** u^{x+1}

e p^{9} **f** p^{3} **g** y^{7} **h** $24s^{8}t^{-2}$

i $2p^{14}$ **j** c^{3z+1} **k** $\dfrac{36}{m^{5}}$ **l** $21y^{3}$

3 $15j^{14} \div \left(3j^{7}\right) = 5j^{7}$

4

a $\dfrac{a^{3}}{b^{3}}$ **b** $x^{7}y^{7}$ **c** $16x^{4}$ **d** $\dfrac{x^{2}y^{2}}{25z^{2}}$

5 2^{16p}

6

a Multiply the exponents **b** Add the exponents

c Subtract the exponents **d** Subtract the exponents

7

a x^{3} **b** u^{3x+3} **c** p^{8} **d** p^{4}

e x^{8} **f** $32y^{14}$ **g** $1024y^{11}$ **h** $108y^{10}$

i $\dfrac{1}{u^{10}}$ **j** $108m^{7}$ **k** $-135m^{3}$ **l** t^{13}

m t^{24} **n** u^{2x+8}

Zero and negative exponents

8

a 1 **b** 8 **c** 1 **d** 1

e -11 **f** 121

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9 mn^{-4}

10

a $\dfrac{4}{a^2}$ **b** $\dfrac{1}{3a}$ **c** $\dfrac{6y^4}{x^3}$ **d** $\dfrac{1}{m^5}$

e $\dfrac{p^4}{m^5n^4}$ **f** $6y^4$ **g** $3x^7$ **h** $\dfrac{3}{x^4}$

11

a $\left(\dfrac{b}{a}\right)^5$ **b** $\dfrac{256}{m^{40}}$ **c** $\dfrac{m^{24}}{64}$ **d** $\dfrac{1}{125y^9}$

e $\dfrac{y^4}{x^2}$ **f** $\dfrac{b^{15}}{a^{15}}$ **g** $\dfrac{81}{16h^4}$ **h** $\dfrac{1}{m^{16}}$

12

a $\dfrac{10}{q^7}$ **b** $\dfrac{1}{8q^{10}}$ **c** $\dfrac{5z}{x^2y^7}$ **d** 2^{20}

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